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FIFTH GENERATION SYSTEM QUALITY OF SERVICE USING MEDICAL STATES

Abstract

A communication device can determine information associated with a medical state of a patient associated with the communication device. The communication device can further determine a priority of data associated with the patient based on the information. The communication device can further communicate data associated with the patient using the priority.

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Background/Summary

INTRODUCTION

[0001] The present disclosure is related to wireless communication systems and more particularly to using a medical state of a patient to determine a quality of service level used for data connectivity of medical data associated with the patient.

BACKGROUND

[0002] FIG. 1 illustrates an example of a new radio (“NR”) network (e.g., a 5th Generation (“5G”) network) including a 5G core (“5GC”) network **130**, network nodes **120a-b** (e.g., 5G base station (“gNB”)), and multiple communication devices **110** (also referred to as user equipment (“UE”)).

[0003] Cellular technologies, such as 5G, are increasingly used for healthcare. As with all communications systems, a 5G system (“5GS”) may be resource constrained during peak traffic events (e.g., system failure events or natural catastrophes), in which case traffic prioritization is necessary to continue to deliver important traffic. This traffic prioritization can be performed by the 5GS quality of service (“QoS”) functions.

SUMMARY

[0004] In some embodiments, a method of operating a communication device in a communications network that includes a network node is provided. The method includes determining information associated with a medical state of a patient associated with the communication device. The method further includes determining a priority of data associated with the patient based on the information. The method further includes communicating data associated with the patient using the priority.

[0005] In other embodiments, a method of operating a network node in a communications network that includes a communication device is provided. The method includes receiving information associated with a medical state of a patient, the patient being associated with the communication device. The method further includes determining a priority of data associated with the patient based on the information. The method further includes prioritizing data associated with the patient based on the priority of the data.

[0006] In other embodiments, a communication device, network node, computer program, computer program product, or non-transitory computer-readable medium is provided for performing one of the methods above.

[0007] In some embodiments, prioritizing data associated with a patient based on their medical state can reduce the risk of the patient's vital medical data being lost by the wireless network, and thereby going unnoticed by medical staff.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] The accompanying drawings, which are included to provide a further understanding of the disclosure and are incorporated in and constitute a part of this application, illustrate certain non-limiting embodiments of inventive concepts. In the drawings:

[0009] FIG. 1 is a schematic diagram illustrating an example of a 5th generation (“5G”) network;

[0010] FIG. 2 is a block diagram illustrating an example of a 5G system (“5GS”) Quality of Service (“QoS”) architecture;

[0011] FIG. 3 is a schematic diagram illustrating an example of 5GS connectivity concepts for QoS;

[0012] FIG. 4 is a table illustrating an example of a scoring system for a set of medical measurements;

[0013] FIG. 5 is a table illustrating an example of a clinical risk and response associated with a score from the scoring system in FIG. 4;

[0014] FIG. 6 is a schematic diagram illustrating an example of a framework for a 5GS QoS using medical states in accordance with some embodiments;

[0015] FIG. 7 is a flow chart illustrating an example of operations performed by a remote UE in accordance with some embodiments;

[0016] FIG. 8 is a flow chart illustrating an example of operations performed by a relay UE in accordance with some embodiments;

[0017] FIG. 9 is a block diagram of a communication system in accordance with some embodiments;

[0018] FIG. 10 is a block diagram of a user equipment in accordance with some embodiments

[0019] FIG. 11 is a block diagram of a network node in accordance with some embodiments;

[0020] FIG. 12 is a block diagram of a host computer communicating with a user equipment in accordance with some embodiments;

[0021] FIG. 13 is a block diagram of a virtualization environment in accordance with some embodiments; and

[0022] FIG. 14 is a block diagram of a host computer communicating via a base station with a user equipment over a partially wireless connection in accordance with some embodiments.

DETAILED DESCRIPTION

[0023] Some of the embodiments contemplated herein will now be described more fully with reference to the accompanying drawings. Embodiments are provided by way of example to convey the scope of the subject matter to those skilled in the art, in which examples of embodiments of inventive concepts are shown. Inventive concepts may, however, be embodied in many different forms and should not be construed as limited to the embodiments set forth herein. Rather, these embodiments are provided so that this disclosure will be thorough and complete, and will fully convey the scope of present inventive concepts to those skilled in the art. It should also be noted that these embodiments are not mutually exclusive. Components from one embodiment may be tacitly assumed to be present/used in another embodiment.

[0024] A 5th Generation system (“5GS”) used in a hospital for example may set Quality of Service (“QoS”) levels based on the type of traffic (e.g., patient measurements of vital signs or medical imaging scans), but that can result in unwanted and detrimental effects: one example would be that a relatively healthy person's pulse measurement is prioritized over a critically ill patient's blood pressure measurement. Losing the latter data during a high data peak time period may result in the medical doctors not detecting a patient going into a critical condition, that may be fatal. There are no solutions that consider the patients' medical conditions to define, set, and change (modify) the QoS levels in 5GS. FIG. 2 illustrates an example of a 5GS that offers QoS across the 5GS.

[0025] QoS includes requirements on all the aspects of connection impairments, such as service response time, loss of data and service, delays, etc. QoS offers the ability to provide different priority to different applications, users, or data flows, or to guarantee a certain level of performance to a data flow.

[0026] In 5G new radio (“NR”), QoS is enforced at the QoS flow level. Packet Detection Rules (“PDRs”) are used by the UPF to classify packets for QoS flow marking using a QoS Flow Identifier (“QFI”). The 5G QoS flows are mapped in the Access Network (“AN”) to Data Radio Bearers (“DRBs”) unlike in 4th Generation (“4G”) where mapping is one to one between Evolved Packet Core (“EPC”) and Radio Bearers. 5G QoS architecture supports the following QoS flow types: Guaranteed Bit Rate (“GBR”) QoS flow; non-GBR QoS flow; and Delay Critical QoS flow. The GBR QoS flow requires a guaranteed flow bit rate. The non-GBR QoS flow does not require guaranteed flow bit rate. The Delay Critical QoS flow is for mission critical guaranteed flow bit rate.

[0027] The QoS architecture in 5G, 5G radio access network (“RAN”) is connected to 5G Core as it is depicted in FIG. 3. The 5G Core establishes one or more protocol data unit (“PDU”) sessions **302** for each UE **110**. The 5G-RAN establishes at least one DRB **304a** together with the PDU Session **302** and additional DRB(s) **304b** for QoS flow(s) **306a-c** of that PDU session **302** can be subsequently configured for each UE **110**. The 5G-RAN maps packets belonging to different PDU

sessions **302** to different DRBs **304a-b**. The non-access stratum (“NAS”) level packet filters in the UE **110** and in the 5GC associate uplink (“UL”) and downlink (“DL”) packets with QoS flows. The Access Stratum (“AS”) level mapping rules in the UE **110** and in the 5G-RAN associate UL and DL QoS flows with DRBs **304a-b**.

[0028] The 5G RAN (e.g., gNB **120**) and the 5G Core (e.g., UPF **260**) ensure quality of service (e.g., reliability and target delay) by mapping packets to appropriate QoS flows **306a-c** and DRBs **304a-b**. Hence there is a 2-step mapping of IP-flows to QoS flows (NAS) and from QoS flows **306a-c** to DRBs **304a-b** (Access Stratum).

[0029] In some examples, the QoS flow is identified by QFI within PDU session **302**. This QFI is carried in an encapsulation header over N3. For each UE, 5GC establishes one or more PDU sessions and 5G-RAN establishes at least one DRB together with PDU session. Additional DRBs are configured for QoS flows of that PDU session consecutively. 5G-RAN maps packets which belong to the different PDU sessions to different DRBs. At the NAS level, a QoS flow is characterized by a QoS profile provided by 5GC to 5G-RAN and QoS rule(s) provided by 5GC to the UE. A QoS flow may either be GBR or Non-GBR depending on its profile. The QoS profile is used by gNB to determine the treatment on the radio interface. QoS rules dictate the mapping between uplink User Plane traffic and QoS flows to the UE.

[0030] At the AS level, the DRB defines the packet treatment on the radio interface (Uu). A DRB serves packets with the same packet forwarding treatment. The QoS flow to DRB mapping by gNB is based on QFI and the associated QoS profiles (i.e., QoS parameters and QoS characteristics). Separate DRBs may be established for QoS flows requiring different packet forwarding treatment, or several QoS flows belonging to the same PDU session can be multiplexed in the same DRB.

[0031] A 5G Network can provide the UE, one or more QoS flow descriptions associated with a PDU session during the PDU session establishment or at the PDU session modification. Each QoS flow includes the following details: a 5G QoS Identifier (“5QI”); and an Allocation and Retention Priority (“ARP”). In case of a GBR QoS flow, the QoS flow can also include a Guaranteed Flow Bit Rate (“GFBR”) for both uplink and downlink; a Maximum Flow Bit Rate (“MFBR”) for both uplink and downlink; a Maximum Packet Loss Rate for both uplink and downlink; a Delay Critical Resource Type; and a Notification Control. In case of Non-GBR QoS flow, the QoS flow can include a Reflective QoS Attribute (“RQA”); a Session-aggregate maximum bit rate (“AMBR”); and a UE-AMBR.

[0032] 5G QoS characteristics describe the packet forwarding treatment that a QoS flow receives edge-to-edge between the UE and the UPF in terms of the following performance characteristics: Resource Type (GBR, Delay critical GBR or Non-GBR); Priority Level; Packet Delay Budget; Packet Error Rate; Averaging window (for GBR and Delay-critical GBR resource type only); and Maximum Data Burst Volume (for Delay-critical GBR resource type only).

[0033] The 5G QoS characteristics should be understood as guidelines for setting node specific parameters for each QoS flow e.g., for 3GPP radio access link layer protocol configurations. Standardized or pre-configured 5G QoS characteristics are indicated through the 5QI value, and are not signaled on any interface, unless certain 5G QoS characteristics are modified.

[0034] In this example of a 5GC, there is only a single user plane function (“UPF”) for transport of data between the gNB and the core. DRBs on air interface can have one-to-many relationship with the general packet radio service (“GPRS”) tunnelling protocol (“GTP-U”) tunnel on N3 interface at UPF. Each QoS flow is mapped to a single GTP-U tunnel at N3 interface. gNB may map individual QoS flows to one more DRBs. A PDU session may contain multiple QoS flows and several DRBs but only a single N3 GTP-U tunnel. A DRB may transport one or more QoS flows. The QFI that identifies the flow is carried in an extension header on N3 in the GTP-U protocol, using DL and UL PDU session information frames. The DL and UL PDU session information frame includes a QFI field for each packet. The DL PDU session information frame includes the Reflective QoS Indicator (“RQI”) field to indicate whether the user plane reflective QoS is to be activated or not.

This is only applicable if reflective QoS is activated.

[0035] There currently exist certain challenges. There are no solutions that consider the patients' medical conditions and other patient attributes to define, set and change (modify) the QoS levels in 5GS.

[0036] Certain aspects of the disclosure and their embodiments may provide solutions to these or other challenges.

[0037] Various embodiments described herein allow a patient's overall healthcare traffic to be prioritized, which may come from different sources, or specific sub streams (such as video), based on the medical state of that patient. In some examples, the traffic associated with a patient is prioritized based on measurements of the patient's vital signs (measurements of the patient's most basic functions). The four main vital signs routinely monitored by medical professionals and health care providers include the following: Body temperature; Heart rate (Pulse); Respiratory rate; and Blood pressure. Other vital signs can include, for example, Oxygen saturation (as measured by pulse oximetry) and Blood Glucose level.

[0038] In some embodiments, these vital signs are used to determine an aggregate numerical value to represent a patient's condition. For example, an aggregate numerical value associated with a patient's medical state can be determined using the National Early Warning Score ("NEWS") system by the United Kingdom's National Health Service ("NHS"), first produced in 2012 and updated (as NEWS2) in December 2017, which advocates a system to standardize the assessment and response to acute illness. The NEWS is based on a simple aggregate scoring system in which a score is allocated to physiological measurements, already recorded in routine practice, when patients present to, or are being monitored in a hospital. Six physiological parameters form the basis of the scoring system: respiration rate; oxygen saturation; systolic blood pressure; pulse rate; level of consciousness or new confusion; and temperature. Each of these parameters can be measured and given a score, for example, according to the chart in FIG. 4. From the scoring system the overall clinical risk and response can be determined, as illustrated in FIG. 5. Systems like NEWS2 or other process and procedures can also be used to determine the patient's overall medical state.

[0039] For the USA, a wide range of terms may be used to describe a patient's condition. The American Hospital Association advises physicians to use the following one-word conditions in describing a patient's condition to those inquiring: Undetermined; Good; Fair; Serious; Critical; and Dead. Undetermined is used to refer to a patient awaiting physician and/or assessment. Good is used to refer to a patient whose vital signs are stable and within normal limits; patient is conscious and comfortable; and indicators are excellent. Fair is used to refer to a patient whose vital signs are stable and within normal limits; patient is conscious, but may be uncomfortable; indicators are favorable. Serious is used to refer to a patient whose vital signs may be unstable and not within normal limits; patient is seriously ill; and indicators are questionable. Critical is used to refer to a patient whose vital signs are unstable and not within normal limits; patient may be unconscious; and indicators are unfavorable. Dead is used to refer to a patient whose vital signs have ceased; and patient has died.

[0040] In addition to a patient's medical states there may be other relevant patient attributes that are inputs to how the healthcare data should be prioritized. In some examples, a patient's medical history, condition (e.g., age, pre-existing conditions), or type of illness can be used as inputs as they can indicate whether the patient has a significant risk of deteriorating. In additional or alternative examples, economic considerations (e.g., the patient's insurance) can be used as input to lead to differentiated treatment. In additional or alternative examples, a patient's status in society (e.g., leader of government, industry, VIP) can be used to determine how to prioritize data associated with a patient.

[0041] In some embodiments, the overall data traffic priority from the above is used with a system interfacing the 5G core, as illustrated in FIG. 6. The 5GS illustrated in FIG. 6 includes a Network

Slice Selection Function (“NSSF”) **202**, Network Slice Authentication and Authorization Function (“NSSAF”) **204**, Authentication Server Function (“AUSF”) **208**, Unified Data Management (“UDM”) **206**, Access and mobility Management Function (“AMF”) **240**, Session Management Function (“SMF”) **250**, Policy Control Function (“PCF”) **230**, Application Function (“AF”) **220**, Network Exposure Function (“NEF”) **290**, Healthcare 5GS QoS Interface **295**, UE **110**, RAN **120** (also referred to here as an AN), User Plane Function (“UPF”) **260**, and Data Network (“DN”) **216**.

[0042] The Healthcare 5GS QoS Interface **295** can include (or be communicatively coupled to) a manual or automated system to provide inputs to request traffic priority level using an application programming interface (“API”) exposed from a 5G system via a Healthcare 5GS QoS Interface function. The manual input can include a user interface for healthcare staff (e.g., ambulance staff, physicians, and administrative staff). The automated system can include an application server with the application software for aggregation of medical equipment measuring vital signs, producing NEWS2 scores, AI assisted predictions, and/or updating the QoS profiles based on new medical state data. This function can make API calls to the 5GC Network exposure function (“NEF”) **290** which in turn makes API calls within the 5GC trust domain including to the SMF **250**, which can modify traffic priority via PDU session modification procedure to the UPF **260**, AN **120**, and UE **110** functions, which controls the data plane QoS flows (for example the QoS flow max and guaranteed bit rates, priority levels).

[0043] Certain embodiments may reduce the risk of a medical patient's vital medical data being lost by the wireless network, and thereby going unnoticed by medical staff. Reducing this risk can improve patient care.

[0044] An example implementation of some embodiments is described below.

[0045] First, a Healthcare user equipment (“HUE”) is activated. Patient A (“P-A”) may have recently experienced an acute medical condition. During recovery, there is a risk of sudden emergencies, so a wearable patch (e.g., the HUE is this example) is placed on P-A, which is continuously measuring vital signs and reporting them using 5G connectivity. In this way, medical staff can quickly be informed of emergencies, and using artificial intelligence (“AI”) also get early notifications of emerging acute conditions. P-A may feel well and move freely within and in the vicinity of the hospital, even doing occasional trips home. Thus, the HUE connectivity can make use of both the indoor hospital facility wireless 5G RAN network as well as the outdoor macro 5G RAN network.

[0046] Second, a QoS flow can be defined. Using the Healthcare 5GS QoS Interface (“H5GQI”) (e.g., H5GQI **295** of FIG. 6), the healthcare staff can manually insert P-A's Patient ID for the HUE, and thereby defined a new QFI. In some examples, the Patient ID is mapped to the patch device (HUE), which has an IP address, which together with the IP address of the server handling this data defines the IP flows of this traffic. In additional or alternative examples, the QoS flow is controlled by the SMF and may be preconfigured, or established via the PDU Establishment procedure or the PDU Session Modification procedure. This QFI is characterized by QoS profile provided by 5GC SMF to 5G-RAN (e.g., used by the gNB) and QoS rule(s) provided by 5GC SMF to the UE.

[0047] Predefined by the 5GC AMF in the gNB (or provided by the AMF over the N2 interface), the QoS flow is using a profile called Medical Vital Signs (5GI number NN). This QoS profile has: an Allocation and Retention Priority (“ARP”) of initially 5, which is based on P-A's age, gender, prior history, and current condition; Guaranteed Flow Bit Rate (“GFBR”) with an uplink of 0.5 Mbps and downlink of 0.1 Mbps; a Maximum Flow Bit Rate (“MFBR”) with uplink and downlink of 10 Mbps; a Maximum Packet Loss Rate with an uplink and downlink of 10.sup.-5 packets within 100 ms; a Delay Critical Resource Type of Delay critical type with a max delay defined of 20 ms; and a Notification Control.

[0048] The QoS rule called Medical Vital Signs is used by the HUE. This QoS rule is mirroring the QoS profile above, and is provided by the SMF to the HUE via the AMF over the N1 reference point.

[0049] In addition to the QoS handling defined above, the vital signs data from P-A is mapped to a Network slice (identified by a Network Slice Selection Assistance Information (“NSSAI”)) being mapped to the PDU session of the above QoS flow. The Network slice may for example be “Healthcare Provider X Patient Observability data”, carrying many patient QoS flows. This Network slice then connects to the data network with the data servers containing the application software using P-A's medical vital signs data.

[0050] Healthcare 5GS QoS Interface traffic QoS flow can be part of the Medical Vital Signs Network slice using the Healthcare 5GS QoS Interface.

[0051] Third, vital signs thresholds and automatic network priority adjustments are made. Using FIGS. 4-5, PA's HUE reports (via the Healthcare 5GS QoS Interface 295 in FIG. 6 and/or an Automated Medical System communicatively coupled to the Healthcare 5GS QoS Interface 295) that the vital signs are degrading resulting in that one or more thresholds are passed. The Automated Medical System then determines that the QoS profiles and rules need to be improved with a better value (e.g., 3) of the ARP and potentially other parameters as well. Using the H5GQI interface to the NEF, the SMF is using a PDU Session Modification procedure to change the QoS parameters of P-A's QoS flow.

[0052] The intended result can be that PA's data is prioritized over other data including other, less critical patients' data, in case of network resource contention.

[0053] ARP is described further below.

[0054] The QoS parameter ARP can include information about the priority level, the pre-emption capability and the pre-emption vulnerability. The ARP priority level defines the relative importance of a resource request to allow in deciding whether a new QoS flow may be accepted or needs to be rejected in the case of resource limitations (typically used for admission control of GBR traffic). It may also be used to decide which existing QoS flow to pre-empt during resource limitations.

[0055] In some embodiments, the range of the ARP priority level is 1 to 15 with 1 as the highest level of priority.

[0056] In additional or alternative embodiments, the ARP priority levels 1-8 should only be assigned to resources for services that are authorized to receive prioritized treatment within an operator domain (e.g., that are authorized by the serving network).

[0057] In additional or alternative embodiments, the ARP priority levels 9-15 may be assigned to resources that are authorized by the home network and thus applicable when a UE is roaming.

[0058] In additional or alternative embodiments, the ARP pre-emption capability defines whether a service data flow may get resources that were already assigned to another service data flow with a lower level of priority (i.e., a higher ARP priority level value).

[0059] In additional or alternative embodiments, the ARP pre-emption capability and the ARP pre-emption vulnerability shall be either set to ‘enabled’ or ‘disabled.’

[0060] In additional or alternative embodiments, the ARP pre-emption vulnerability defines whether a service data flow may lose the resources assigned to it in order to admit a service data flow with higher ARP priority level.

[0061] In additional or alternative embodiments, the ARP pre-emption vulnerability of the QoS flow which the default QoS rule is associated with should be set appropriately to minimize the risk of unnecessary release of this QoS flow.

[0062] In the description that follows, while the communication device may be any of UE 110, 912A-D, 1000, hardware 1304, or virtual machine 1308A, 1308B, the communication device 1000 shall be used to describe the functionality of the operations of the communication device.

Operations of the communication device 1000 (implemented using the structure of FIG. 10) will now be discussed with reference to the flow chart of FIG. 7 according to some embodiments of inventive concepts. For example, modules may be stored in memory 1010 of FIG. 10, and these modules may provide instructions so that when the instructions of a module are executed by respective communication device processing circuitry 1002, processing circuitry 1002 performs

respective operations of the flow chart.

[0063] FIG. 7 illustrates an example of operations performed by a communication device in a communications network that includes a network node. In some embodiments, the network node includes at least one of: a radio access network, RAN, node; and a core network, CN, node. In additional or alternative embodiments, the communication device includes at least one of: a healthcare user equipment, HUE; a wearable device; and a medical sensor.

[0064] At block **710**, processing circuitry **1002** determines information associated with a medical state of a patient. In some embodiments, the information includes at least one of: a respiration rate of the patient; an oxygen saturation of the patient; a systolic blood pressure of the patient; a pulse rate of the patient; a level of consciousness of the patient; a temperature of the patient; an aggregate numerical value indicating a medical state of the patient; and a term indicating a severity of the medical state of the patient.

[0065] At block **720**, processing circuitry **1002** determines a priority of data associated with the patient based on the information. In some embodiments, determining the priority of the data associated with the patient based on the information includes transmitting the information to the network node; and responsive to transmitting the information, receiving an indication of a priority of the data. In additional or alternative embodiments, receiving the indication of the priority of the data includes receiving a Quality of Service Flow Identifier, QFI, associated with a Quality of Service, QoS, profile associated with the data. In some examples, the QoS profile includes an indication of at least one of: an allocation and retention priority, ARP; a guaranteed flow bit rate, GFBR; a maximum flow bit rate, MFBR; a maximum packet loss rate; a delay critical resource type; and a notification control.

[0066] In some embodiments, determining the priority of the data associated with the patient based on the information includes determining at least one of: whether the medical state of the patient exceeds a first threshold level; whether the medical state of the patient has fallen below a second threshold level; whether the medical state of the patient exceeds a threshold range; and whether a rate of change in the medical state of the patient exceeds a threshold rate. In some examples, determining that the medical state of the patient exceeds the first threshold includes determining that a patient's temperature exceeds a healthy temperature. In additional or alternative examples, determining that the medical state of the patient has fallen below the second threshold level includes determining that the patient's oxygen level has fallen below a healthy oxygen level. In additional or alternative examples, determining that the medical state of the patient exceeds the threshold range includes determining that the patient's heart rate is above or below a healthy heart rate range. In additional or alternative examples, determining that the rate of change in the medical state of the patient exceeds a threshold rate includes determining that a patient's NEW score has increased at such a high rate (above the third threshold level) that the patient is considered to be quickly deteriorating and therefore the patient's data is given a higher priority.

[0067] At block **730**, processing circuitry **1002** communicates, via communication interface **1012**, data associated with the patient using the priority. In some embodiments, the patient includes a first patient of a plurality of patients associated with the communication device. Communicating the data associated with the patient includes prioritizing the data associated with the first patient over data associated with a second patient of the plurality of patients.

[0068] In some embodiments, communicating the data associated with the patient includes transmitting a message including the data and the QFI in the header.

[0069] Various operations of FIG. 7 may be optional with respect to some embodiments.

[0070] In the description that follows, while the network node may be any of RAN **120**, UPF **260**, AMF **240**, SMF **250**, NEF **290**, H5GQI **295**, CN Node **908**, Network Node **910A-B**, **1100**, hardware **1304**, or virtual machine **1308A**, **1308B**, the network node **1100** shall be used to describe the functionality of the operations of the network node. Operations of the network node **1100** (implemented using the structure of FIG. 11) will now be discussed with reference to the flow chart

of FIG. 8 according to some embodiments of inventive concepts. For example, modules may be stored in memory **1104** of FIG. 11, and these modules may provide instructions so that when the instructions of a module are executed by respective network node processing circuitry **1102**, processing circuitry **1102** performs respective operations of the flow chart.

[0071] FIG. 8 illustrates an example of operations performed by a network node in a communications network that includes a communication device. In some embodiments, the network node includes at least one of: a radio access network, RAN, node; and a core network, CN, node. The communication device includes at least one of: a healthcare user equipment, HUE; a wearable device; and a medical sensor.

[0072] At block **810**, processing circuitry **1102** receives, via communication interface **1106**, information associated with a medical state of a patient. In some embodiments, receiving the information includes receiving a request from the communication device to determine the priority of the data associated with the patient. In additional or alternative embodiments, receiving the information includes receiving at least one of: a respiration rate of the patient; an oxygen saturation of the patient; a systolic blood pressure of the patient; a pulse rate of the patient; a level of consciousness of the patient; a temperature of the patient; an aggregate numerical value indicating a medical state of the patient; and a term indicating a severity of the medical state of the patient.

[0073] At block **820**, processing circuitry **1102** determines a priority of data associated with the patient based on the information. In some embodiments, determining the priority of the data associated with the patient includes determining the priority of the data based on the medical state exceeding a threshold.

[0074] In some embodiments, the information includes current information indicating a current medical state of the patient. Determining the priority of the data associated with the patient includes determining the priority of the data based on a change between the current information and previous information indicating a previous medical state of the patient. In additional or alternative embodiments, determining the priority of the data associated with patient further includes determining the priority of the data associated with the patient based on an amount of time between receiving the current information and receiving the previous information. In some examples, a patient may have a NEW score of 2 at a first time (e.g., right after a surgery) and then the patient may have a NEW score of 6 at a second time (e.g., 1 hour after the surgery). Although a new score of 6 may only be considered a medium risk (such that the patient's data is given a medium priority), the fact that the patient's medical state has dropped (gone up in NEW score) so quickly may be an indicator that the patient's medical state is rapidly deteriorating. Accordingly, by observing the change in the patient's medical state, the processing circuitry **1102** may determine to give greater priority to the data associated with the patient.

[0075] In some embodiments, determining the priority of the data includes creating or modifying a Quality of Service, QoS, profile associated with the data. In some examples, the QoS profile includes an indication of at least one of: an allocation and retention priority, ARP; a guaranteed flow bit rate, GFBR; a maximum flow bit rate, MFBR; a maximum packet loss rate; a delay critical resource type; and a notification control.

[0076] At block **830**, processing circuitry **1102** prioritizes data associated with the patient based on the priority of the data. In some embodiments, prioritizing the data associated with the patient includes transmitting an indication of a QoS Flow Identifier, QFI, associated with the QoS profile to the communication device with instructions to include the QFI in a header of messages that include the data.

[0077] In some embodiments, prioritizing the data associated with the patient includes: receiving a plurality of data including data associated with the patient; and communicating the data associated with the patient before other data of the plurality of data based on the priority of the data associated with the patient.

[0078] In some embodiments, prioritizing the data associated with the patient includes allocating

more network resources for the data associated with the patient than for other data being communicated on the communications network.

[0079] Various operations of FIG. 8 may be optional with respect to some embodiments.

[0080] FIG. 9 shows an example of a communication system 900 in accordance with some embodiments.

[0081] In the example, the communication system 900 includes a telecommunication network 902 that includes an access network 904, such as a radio access network (RAN), and a core network 906, which includes one or more core network nodes 908. The access network 904 includes one or more access network nodes, such as network nodes 910a and 910b (one or more of which may be generally referred to as network nodes 910), or any other similar 3.sup.rd Generation Partnership Project (3GPP) access node or non-3GPP access point. The network nodes 910 facilitate direct or indirect connection of user equipment (UE), such as by connecting UEs 912a, 912b, 912c, and 912d (one or more of which may be generally referred to as UEs 912) to the core network 906 over one or more wireless connections.

[0082] Example wireless communications over a wireless connection include transmitting and/or receiving wireless signals using electromagnetic waves, radio waves, infrared waves, and/or other types of signals suitable for conveying information without the use of wires, cables, or other material conductors. Moreover, in different embodiments, the communication system 900 may include any number of wired or wireless networks, network nodes, UEs, and/or any other components or systems that may facilitate or participate in the communication of data and/or signals whether via wired or wireless connections. The communication system 900 may include and/or interface with any type of communication, telecommunication, data, cellular, radio network, and/or other similar type of system.

[0083] The UEs 912 may be any of a wide variety of communication devices, including wireless devices arranged, configured, and/or operable to communicate wirelessly with the network nodes 910 and other communication devices. Similarly, the network nodes 910 are arranged, capable, configured, and/or operable to communicate directly or indirectly with the UEs 912 and/or with other network nodes or equipment in the telecommunication network 902 to enable and/or provide network access, such as wireless network access, and/or to perform other functions, such as administration in the telecommunication network 902.

[0084] In the depicted example, the core network 906 connects the network nodes 910 to one or more hosts, such as host 916. These connections may be direct or indirect via one or more intermediary networks or devices. In other examples, network nodes may be directly coupled to hosts. The core network 906 includes one more core network nodes (e.g., core network node 908) that are structured with hardware and software components. Features of these components may be substantially similar to those described with respect to the UEs, network nodes, and/or hosts, such that the descriptions thereof are generally applicable to the corresponding components of the core network node 908. Example core network nodes include functions of one or more of a Mobile Switching Center (MSC), Mobility Management Entity (MME), Home Subscriber Server (HSS), Access and Mobility Management Function (AMF), Session Management Function (SMF), Authentication Server Function (AUSF), Subscription Identifier De-concealing function (SIDF), Unified Data Management (UDM), Security Edge Protection Proxy (SEPP), Network Exposure Function (NEF), and/or a User Plane Function (UPF).

[0085] The host 916 may be under the ownership or control of a service provider other than an operator or provider of the access network 904 and/or the telecommunication network 902, and may be operated by the service provider or on behalf of the service provider. The host 916 may host a variety of applications to provide one or more service. Examples of such applications include live and pre-recorded audio/video content, data collection services such as retrieving and compiling data on various ambient conditions detected by a plurality of UEs, analytics functionality, social media, functions for controlling or otherwise interacting with remote devices, functions for an

alarm and surveillance center, or any other such function performed by a server.

[0086] As a whole, the communication system **900** of FIG. **9** enables connectivity between the UEs, network nodes, and hosts. In that sense, the communication system may be configured to operate according to predefined rules or procedures, such as specific standards that include, but are not limited to: Global System for Mobile Communications (GSM); Universal Mobile Telecommunications System (UMTS); Long Term Evolution (LTE), and/or other suitable 2G, 3G, 4G, 5G standards, or any applicable future generation standard (e.g., 6G); wireless local area network (WLAN) standards, such as the Institute of Electrical and Electronics Engineers (IEEE) 802.11 standards (WiFi); and/or any other appropriate wireless communication standard, such as the Worldwide Interoperability for Microwave Access (WiMax), Bluetooth, Z-Wave, Near Field Communication (NFC) ZigBee, LiFi, and/or any low-power wide-area network (LPWAN) standards such as LoRa and Sigfox.

[0087] In some examples, the telecommunication network **902** is a cellular network that implements 3GPP standardized features. Accordingly, the telecommunications network **902** may support network slicing to provide different logical networks to different devices that are connected to the telecommunication network **902**. For example, the telecommunications network **902** may provide Ultra Reliable Low Latency Communication (URLLC) services to some UEs, while providing Enhanced Mobile Broadband (eMBB) services to other UEs, and/or Massive Machine Type Communication (mMTC)/Massive IoT services to yet further UEs.

[0088] In some examples, the UEs **912** are configured to transmit and/or receive information without direct human interaction. For instance, a UE may be designed to transmit information to the access network **904** on a predetermined schedule, when triggered by an internal or external event, or in response to requests from the access network **904**. Additionally, a UE may be configured for operating in single- or multi-RAT or multi-standard mode. For example, a UE may operate with any one or combination of Wi-Fi, NR (New Radio) and LTE, i.e., being configured for multi-radio dual connectivity (MR-DC), such as E-UTRAN (Evolved-UMTS Terrestrial Radio Access Network) New Radio-Dual Connectivity (EN-DC).

[0089] In the example, the hub **914** communicates with the access network **904** to facilitate indirect communication between one or more UEs (e.g., UE **912c** and/or **912d**) and network nodes (e.g., network node **910b**). In some examples, the hub **914** may be a controller, router, content source and analytics, or any of the other communication devices described herein regarding UEs. For example, the hub **914** may be a broadband router enabling access to the core network **906** for the UEs. As another example, the hub **914** may be a controller that sends commands or instructions to one or more actuators in the UEs. Commands or instructions may be received from the UEs, network nodes **910**, or by executable code, script, process, or other instructions in the hub **914**. As another example, the hub **914** may be a data collector that acts as temporary storage for UE data and, in some embodiments, may perform analysis or other processing of the data. As another example, the hub **914** may be a content source. For example, for a UE that is a VR headset, display, loudspeaker or other media delivery device, the hub **914** may retrieve VR assets, video, audio, or other media or data related to sensory information via a network node, which the hub **914** then provides to the UE either directly, after performing local processing, and/or after adding additional local content. In still another example, the hub **914** acts as a proxy server or orchestrator for the UEs, in particular in if one or more of the UEs are low energy IoT devices.

[0090] The hub **914** may have a constant/persistent or intermittent connection to the network node **910b**. The hub **914** may also allow for a different communication scheme and/or schedule between the hub **914** and UEs (e.g., UE **912c** and/or **912d**), and between the hub **914** and the core network **906**. In other examples, the hub **914** is connected to the core network **906** and/or one or more UEs via a wired connection. Moreover, the hub **914** may be configured to connect to an M2M service provider over the access network **904** and/or to another UE over a direct connection. In some scenarios, UEs may establish a wireless connection with the network nodes **910** while still

connected via the hub **914** via a wired or wireless connection. In some embodiments, the hub **914** may be a dedicated hub—that is, a hub whose primary function is to route communications to/from the UEs from/to the network node **910b**. In other embodiments, the hub **914** may be a non-dedicated hub—that is, a device which is capable of operating to route communications between the UEs and network node **910b**, but which is additionally capable of operating as a communication start and/or end point for certain data channels.

[0091] FIG. **10** shows a UE **1000** in accordance with some embodiments. As used herein, a UE refers to a device capable, configured, arranged and/or operable to communicate wirelessly with network nodes and/or other UEs. Examples of a UE include, but are not limited to, a smart phone, mobile phone, cell phone, voice over IP (VOIP) phone, wireless local loop phone, desktop computer, personal digital assistant (PDA), wireless cameras, gaming console or device, music storage device, playback appliance, wearable terminal device, wireless endpoint, mobile station, tablet, laptop, laptop-embedded equipment (LEE), laptop-mounted equipment (LME), smart device, wireless customer-premise equipment (CPE), vehicle-mounted or vehicle embedded/integrated wireless device, etc. Other examples include any UE identified by the 3rd Generation Partnership Project (3GPP), including a narrow band internet of things (NB-IoT) UE, a machine type communication (MTC) UE, and/or an enhanced MTC (eMTC) UE.

[0092] A UE may support device-to-device (D2D) communication, for example by implementing a 3GPP standard for sidelink communication, Dedicated Short-Range Communication (DSRC), vehicle-to-vehicle (V2V), vehicle-to-infrastructure (V2I), or vehicle-to-everything (V2X). In other examples, a UE may not necessarily have a user in the sense of a human user who owns and/or operates the relevant device. Instead, a UE may represent a device that is intended for sale to, or operation by, a human user but which may not, or which may not initially, be associated with a specific human user (e.g., a smart sprinkler controller). Alternatively, a UE may represent a device that is not intended for sale to, or operation by, an end user but which may be associated with or operated for the benefit of a user (e.g., a smart power meter).

[0093] The UE **1000** includes processing circuitry **1002** that is operatively coupled via a bus **1004** to an input/output interface **1006**, a power source **1008**, a memory **1010**, a communication interface **1012**, and/or any other component, or any combination thereof. Certain UEs may utilize all or a subset of the components shown in FIG. **10**. The level of integration between the components may vary from one UE to another UE. Further, certain UEs may contain multiple instances of a component, such as multiple processors, memories, transceivers, transmitters, receivers, etc.

[0094] The processing circuitry **1002** is configured to process instructions and data and may be configured to implement any sequential state machine operative to execute instructions stored as machine-readable computer programs in the memory **1010**. The processing circuitry **1002** may be implemented as one or more hardware-implemented state machines (e.g., in discrete logic, field-programmable gate arrays (FPGAs), application specific integrated circuits (ASICs), etc.); programmable logic together with appropriate firmware; one or more stored computer programs, general-purpose processors, such as a microprocessor or digital signal processor (DSP), together with appropriate software; or any combination of the above. For example, the processing circuitry **1002** may include multiple central processing units (CPUs).

[0095] In the example, the input/output interface **1006** may be configured to provide an interface or interfaces to an input device, output device, or one or more input and/or output devices. Examples of an output device include a speaker, a sound card, a video card, a display, a monitor, a printer, an actuator, an emitter, a smartcard, another output device, or any combination thereof. An input device may allow a user to capture information into the UE **1000**. Examples of an input device include a touch-sensitive or presence-sensitive display, a camera (e.g., a digital camera, a digital video camera, a web camera, etc.), a microphone, a sensor, a mouse, a trackball, a directional pad, a trackpad, a scroll wheel, a smartcard, and the like. The presence-sensitive display may include a capacitive or resistive touch sensor to sense input from a user. A sensor may be, for instance, an

accelerometer, a gyroscope, a tilt sensor, a force sensor, a magnetometer, an optical sensor, a proximity sensor, a biometric sensor, etc., or any combination thereof. An output device may use the same type of interface port as an input device. For example, a Universal Serial Bus (USB) port may be used to provide an input device and an output device.

[0096] In some embodiments, the power source **1008** is structured as a battery or battery pack. Other types of power sources, such as an external power source (e.g., an electricity outlet), photovoltaic device, or power cell, may be used. The power source **1008** may further include power circuitry for delivering power from the power source **1008** itself, and/or an external power source, to the various parts of the UE **1000** via input circuitry or an interface such as an electrical power cable. Delivering power may be, for example, for charging of the power source **1008**. Power circuitry may perform any formatting, converting, or other modification to the power from the power source **1008** to make the power suitable for the respective components of the UE **1000** to which power is supplied.

[0097] The memory **1010** may be or be configured to include memory such as random access memory (RAM), read-only memory (ROM), programmable read-only memory (PROM), erasable programmable read-only memory (EPROM), electrically erasable programmable read-only memory (EEPROM), magnetic disks, optical disks, hard disks, removable cartridges, flash drives, and so forth. In one example, the memory **1010** includes one or more application programs **1014**, such as an operating system, web browser application, a widget, gadget engine, or other application, and corresponding data **1016**. The memory **1010** may store, for use by the UE **1000**, any of a variety of various operating systems or combinations of operating systems.

[0098] The memory **1010** may be configured to include a number of physical drive units, such as redundant array of independent disks (RAID), flash memory, USB flash drive, external hard disk drive, thumb drive, pen drive, key drive, high-density digital versatile disc (HD-DVD) optical disc drive, internal hard disk drive, Blu-Ray optical disc drive, holographic digital data storage (HDDS) optical disc drive, external mini-dual in-line memory module (DIMM), synchronous dynamic random access memory (SDRAM), external micro-DIMM SDRAM, smartcard memory such as tamper resistant module in the form of a universal integrated circuit card (UICC) including one or more subscriber identity modules (SIMs), such as a USIM and/or ISIM, other memory, or any combination thereof. The UICC may for example be an embedded UICC (eUICC), integrated UICC (iUICC) or a removable UICC commonly known as 'SIM card.' The memory **1010** may allow the UE **1000** to access instructions, application programs and the like, stored on transitory or non-transitory memory media, to off-load data, or to upload data. An article of manufacture, such as one utilizing a communication system may be tangibly embodied as or in the memory **1010**, which may be or comprise a device-readable storage medium.

[0099] The processing circuitry **1002** may be configured to communicate with an access network or other network using the communication interface **1012**. The communication interface **1012** may comprise one or more communication subsystems and may include or be communicatively coupled to an antenna **1022**. The communication interface **1012** may include one or more transceivers used to communicate, such as by communicating with one or more remote transceivers of another device capable of wireless communication (e.g., another UE or a network node in an access network). Each transceiver may include a transmitter **1018** and/or a receiver **1020** appropriate to provide network communications (e.g., optical, electrical, frequency allocations, and so forth). Moreover, the transmitter **1018** and receiver **1020** may be coupled to one or more antennas (e.g., antenna **1022**) and may share circuit components, software or firmware, or alternatively be implemented separately.

[0100] In the illustrated embodiment, communication functions of the communication interface **1012** may include cellular communication, Wi-Fi communication, LPWAN communication, data communication, voice communication, multimedia communication, short-range communications such as Bluetooth, near-field communication, location-based communication such as the use of the

global positioning system (GPS) to determine a location, another like communication function, or any combination thereof.

[0101] Communications may be implemented in according to one or more communication protocols and/or standards, such as IEEE 802.11, Code Division Multiplexing Access (CDMA), Wideband Code Division Multiple Access (WCDMA), GSM, LTE, New Radio (NR), UMTS, WiMax, Ethernet, transmission control protocol/internet protocol (TCP/IP), synchronous optical networking (SONET), Asynchronous Transfer Mode (ATM), QUIC, Hypertext Transfer Protocol (HTTP), and so forth.

[0102] Regardless of the type of sensor, a UE may provide an output of data captured by its sensors, through its communication interface **1012**, via a wireless connection to a network node. Data captured by sensors of a UE can be communicated through a wireless connection to a network node via another UE. The output may be periodic (e.g., once every 15 minutes if it reports the sensed temperature), random (e.g., to even out the load from reporting from several sensors), in response to a triggering event (e.g., when moisture is detected an alert is sent), in response to a request (e.g., a user initiated request), or a continuous stream (e.g., a live video feed of a patient).

[0103] As another example, a UE comprises an actuator, a motor, or a switch, related to a communication interface configured to receive wireless input from a network node via a wireless connection. In response to the received wireless input the states of the actuator, the motor, or the switch may change. For example, the UE may comprise a motor that adjusts the control surfaces or rotors of a drone in flight according to the received input or to a robotic arm performing a medical procedure according to the received input.

[0104] A UE, when in the form of an Internet of Things (IoT) device, may be a device for use in one or more application domains, these domains comprising, but not limited to, city wearable technology, extended industrial application and healthcare. Non-limiting examples of such an IoT device are a device which is or which is embedded in: a connected refrigerator or freezer, a TV, a connected lighting device, an electricity meter, a robot vacuum cleaner, a voice controlled smart speaker, a home security camera, a motion detector, a thermostat, a smoke detector, a door/window sensor, a flood/moisture sensor, an electrical door lock, a connected doorbell, an air conditioning system like a heat pump, an autonomous vehicle, a surveillance system, a weather monitoring device, a vehicle parking monitoring device, an electric vehicle charging station, a smart watch, a fitness tracker, a head-mounted display for Augmented Reality (AR) or Virtual Reality (VR), a wearable for tactile augmentation or sensory enhancement, a water sprinkler, an animal- or item-tracking device, a sensor for monitoring a plant or animal, an industrial robot, an Unmanned Aerial Vehicle (UAV), and any kind of medical device, like a heart rate monitor or a remote controlled surgical robot. A UE in the form of an IoT device comprises circuitry and/or software in dependence of the intended application of the IoT device in addition to other components as described in relation to the UE **1000** shown in FIG. **10**.

[0105] As yet another specific example, in an IoT scenario, a UE may represent a machine or other device that performs monitoring and/or measurements, and transmits the results of such monitoring and/or measurements to another UE and/or a network node. The UE may in this case be an M2M device, which may in a 3GPP context be referred to as an MTC device. As one particular example, the UE may implement the 3GPP NB-IoT standard. In other scenarios, a UE may represent a vehicle, such as a car, a bus, a truck, a ship and an airplane, or other equipment that is capable of monitoring and/or reporting on its operational status or other functions associated with its operation.

[0106] In practice, any number of UEs may be used together with respect to a single use case. For example, a first UE might be or be integrated in a drone and provide the drone's speed information (obtained through a speed sensor) to a second UE that is a remote controller operating the drone. When the user makes changes from the remote controller, the first UE may adjust the throttle on the drone (e.g., by controlling an actuator) to increase or decrease the drone's speed. The first

and/or the second UE can also include more than one of the functionalities described above. For example, a UE might comprise the sensor and the actuator, and handle communication of data for both the speed sensor and the actuators.

[0107] FIG. **11** shows a network node **1100** in accordance with some embodiments. As used herein, network node refers to equipment capable, configured, arranged and/or operable to communicate directly or indirectly with a UE and/or with other network nodes or equipment, in a telecommunication network. Examples of network nodes include, but are not limited to, access points (APs) (e.g., radio access points), base stations (BSs) (e.g., radio base stations, Node Bs, evolved Node Bs (eNBs) and NR NodeBs (gNBs)).

[0108] Base stations may be categorized based on the amount of coverage they provide (or, stated differently, their transmit power level) and so, depending on the provided amount of coverage, may be referred to as femto base stations, pico base stations, micro base stations, or macro base stations. A base station may be a relay node or a relay donor node controlling a relay. A network node may also include one or more (or all) parts of a distributed radio base station such as centralized digital units and/or remote radio units (RRUs), sometimes referred to as Remote Radio Heads (RRHs). Such remote radio units may or may not be integrated with an antenna as an antenna integrated radio. Parts of a distributed radio base station may also be referred to as nodes in a distributed antenna system (DAS).

[0109] Other examples of network nodes include multiple transmission point (multi-TRP) 5G access nodes, multi-standard radio (MSR) equipment such as MSR BSs, network controllers such as radio network controllers (RNCs) or base station controllers (BSCs), base transceiver stations (BTSs), transmission points, transmission nodes, multi-cell/multicast coordination entities (MCEs), Operation and Maintenance (O&M) nodes, Operations Support System (OSS) nodes, Self-Organizing Network (SON) nodes, positioning nodes (e.g., Evolved Serving Mobile Location Centers (E-SMLCs)), and/or Minimization of Drive Tests (MDTs).

[0110] The network node **1100** includes a processing circuitry **1102**, a memory **1104**, a communication interface **1106**, and a power source **1108**. The network node **1100** may be composed of multiple physically separate components (e.g., a NodeB component and a RNC component, or a BTS component and a BSC component, etc.), which may each have their own respective components. In certain scenarios in which the network node **1100** comprises multiple separate components (e.g., BTS and BSC components), one or more of the separate components may be shared among several network nodes. For example, a single RNC may control multiple NodeBs. In such a scenario, each unique NodeB and RNC pair, may in some instances be considered a single separate network node. In some embodiments, the network node **1100** may be configured to support multiple radio access technologies (RATs). In such embodiments, some components may be duplicated (e.g., separate memory **1104** for different RATs) and some components may be reused (e.g., a same antenna **1110** may be shared by different RATs). The network node **1100** may also include multiple sets of the various illustrated components for different wireless technologies integrated into network node **1100**, for example GSM, WCDMA, LTE, NR, WiFi, Zigbee, Z-wave, LoRaWAN, Radio Frequency Identification (RFID) or Bluetooth wireless technologies. These wireless technologies may be integrated into the same or different chip or set of chips and other components within network node **1100**.

[0111] The processing circuitry **1102** may comprise a combination of one or more of a microprocessor, controller, microcontroller, central processing unit, digital signal processor, application-specific integrated circuit, field programmable gate array, or any other suitable computing device, resource, or combination of hardware, software and/or encoded logic operable to provide, either alone or in conjunction with other network node **1100** components, such as the memory **1104**, to provide network node **1100** functionality.

[0112] In some embodiments, the processing circuitry **1102** includes a system on a chip (SOC). In some embodiments, the processing circuitry **1102** includes one or more of radio frequency (RF)

transceiver circuitry **1112** and baseband processing circuitry **1114**. In some embodiments, the radio frequency (RF) transceiver circuitry **1112** and the baseband processing circuitry **1114** may be on separate chips (or sets of chips), boards, or units, such as radio units and digital units. In alternative embodiments, part or all of RF transceiver circuitry **1112** and baseband processing circuitry **1114** may be on the same chip or set of chips, boards, or units.

[0113] The memory **1104** may comprise any form of volatile or non-volatile computer-readable memory including, without limitation, persistent storage, solid-state memory, remotely mounted memory, magnetic media, optical media, random access memory (RAM), read-only memory (ROM), mass storage media (for example, a hard disk), removable storage media (for example, a flash drive, a Compact Disk (CD) or a Digital Video Disk (DVD)), and/or any other volatile or non-volatile, non-transitory device-readable and/or computer-executable memory devices that store information, data, and/or instructions that may be used by the processing circuitry **1102**. The memory **1104** may store any suitable instructions, data, or information, including a computer program, software, an application including one or more of logic, rules, code, tables, and/or other instructions capable of being executed by the processing circuitry **1102** and utilized by the network node **1100**. The memory **1104** may be used to store any calculations made by the processing circuitry **1102** and/or any data received via the communication interface **1106**. In some embodiments, the processing circuitry **1102** and memory **1104** is integrated.

[0114] The communication interface **1106** is used in wired or wireless communication of signaling and/or data between a network node, access network, and/or UE. As illustrated, the communication interface **1106** comprises port(s)/terminal(s) **1116** to send and receive data, for example to and from a network over a wired connection. The communication interface **1106** also includes radio front-end circuitry **1118** that may be coupled to, or in certain embodiments a part of, the antenna **1110**. Radio front-end circuitry **1118** comprises filters **1120** and amplifiers **1122**. The radio front-end circuitry **1118** may be connected to an antenna **1110** and processing circuitry **1102**. The radio front-end circuitry may be configured to condition signals communicated between antenna **1110** and processing circuitry **1102**. The radio front-end circuitry **1118** may receive digital data that is to be sent out to other network nodes or UEs via a wireless connection. The radio front-end circuitry **1118** may convert the digital data into a radio signal having the appropriate channel and bandwidth parameters using a combination of filters **1120** and/or amplifiers **1122**. The radio signal may then be transmitted via the antenna **1110**. Similarly, when receiving data, the antenna **1110** may collect radio signals which are then converted into digital data by the radio front-end circuitry **1118**. The digital data may be passed to the processing circuitry **1102**. In other embodiments, the communication interface may comprise different components and/or different combinations of components.

[0115] In certain alternative embodiments, the network node **1100** does not include separate radio front-end circuitry **1118**, instead, the processing circuitry **1102** includes radio front-end circuitry and is connected to the antenna **1110**. Similarly, in some embodiments, all or some of the RF transceiver circuitry **1112** is part of the communication interface **1106**. In still other embodiments, the communication interface **1106** includes one or more ports or terminals **1116**, the radio front-end circuitry **1118**, and the RF transceiver circuitry **1112**, as part of a radio unit (not shown), and the communication interface **1106** communicates with the baseband processing circuitry **1114**, which is part of a digital unit (not shown).

[0116] The antenna **1110** may include one or more antennas, or antenna arrays, configured to send and/or receive wireless signals. The antenna **1110** may be coupled to the radio front-end circuitry **1118** and may be any type of antenna capable of transmitting and receiving data and/or signals wirelessly. In certain embodiments, the antenna **1110** is separate from the network node **1100** and connectable to the network node **1100** through an interface or port.

[0117] The antenna **1110**, communication interface **1106**, and/or the processing circuitry **1102** may be configured to perform any receiving operations and/or certain obtaining operations described

herein as being performed by the network node. Any information, data and/or signals may be received from a UE, another network node and/or any other network equipment. Similarly, the antenna **1110**, the communication interface **1106**, and/or the processing circuitry **1102** may be configured to perform any transmitting operations described herein as being performed by the network node. Any information, data and/or signals may be transmitted to a UE, another network node and/or any other network equipment.

[0118] The power source **1108** provides power to the various components of network node **1100** in a form suitable for the respective components (e.g., at a voltage and current level needed for each respective component). The power source **1108** may further comprise, or be coupled to, power management circuitry to supply the components of the network node **1100** with power for performing the functionality described herein. For example, the network node **1100** may be connectable to an external power source (e.g., the power grid, an electricity outlet) via an input circuitry or interface such as an electrical cable, whereby the external power source supplies power to power circuitry of the power source **1108**. As a further example, the power source **1108** may comprise a source of power in the form of a battery or battery pack which is connected to, or integrated in, power circuitry. The battery may provide backup power should the external power source fail.

[0119] Embodiments of the network node **1100** may include additional components beyond those shown in FIG. **11** for providing certain aspects of the network node's functionality, including any of the functionality described herein and/or any functionality necessary to support the subject matter described herein. For example, the network node **1100** may include user interface equipment to allow input of information into the network node **1100** and to allow output of information from the network node **1100**. This may allow a user to perform diagnostic, maintenance, repair, and other administrative functions for the network node **1100**.

[0120] FIG. **12** is a block diagram of a host **1200**, which may be an embodiment of the host **916** of FIG. **9**, in accordance with various aspects described herein. As used herein, the host **1200** may be or comprise various combinations hardware and/or software, including a standalone server, a blade server, a cloud-implemented server, a distributed server, a virtual machine, container, or processing resources in a server farm. The host **1200** may provide one or more services to one or more UEs.

[0121] The host **1200** includes processing circuitry **1202** that is operatively coupled via a bus **1204** to an input/output interface **1206**, a network interface **1208**, a power source **1210**, and a memory **1212**. Other components may be included in other embodiments. Features of these components may be substantially similar to those described with respect to the devices of previous figures, such as FIGS. **10-11**, such that the descriptions thereof are generally applicable to the corresponding components of host **1200**.

[0122] The memory **1212** may include one or more computer programs including one or more host application programs **1214** and data **1216**, which may include user data, e.g., data generated by a UE for the host **1200** or data generated by the host **1200** for a UE. Embodiments of the host **1200** may utilize only a subset or all of the components shown. The host application programs **1214** may be implemented in a container-based architecture and may provide support for video codecs (e.g., Versatile Video Coding (VVC), High Efficiency Video Coding (HEVC), Advanced Video Coding (AVC), MPEG, VP9) and audio codecs (e.g., FLAC, Advanced Audio Coding (AAC), MPEG, G.711), including transcoding for multiple different classes, types, or implementations of UEs (e.g., handsets, desktop computers, wearable display systems, heads-up display systems). The host application programs **1214** may also provide for user authentication and licensing checks and may periodically report health, routes, and content availability to a central node, such as a device in or on the edge of a core network. Accordingly, the host **1200** may select and/or indicate a different host for over-the-top services for a UE. The host application programs **1214** may support various protocols, such as the HTTP Live Streaming (HLS) protocol, Real-Time Messaging Protocol (RTMP), Real-Time Streaming Protocol (RTSP), Dynamic Adaptive Streaming over HTTP

(MPEG-DASH), etc.

[0123] FIG. **13** is a block diagram illustrating a virtualization environment **1300** in which functions implemented by some embodiments may be virtualized. In the present context, virtualizing means creating virtual versions of apparatuses or devices which may include virtualizing hardware platforms, storage devices and networking resources. As used herein, virtualization can be applied to any device described herein, or components thereof, and relates to an implementation in which at least a portion of the functionality is implemented as one or more virtual components. Some or all of the functions described herein may be implemented as virtual components executed by one or more virtual machines (VMs) implemented in one or more virtual environments **1300** hosted by one or more of hardware nodes, such as a hardware computing device that operates as a network node, UE, core network node, or host. Further, in embodiments in which the virtual node does not require radio connectivity (e.g., a core network node or host), then the node may be entirely virtualized.

[0124] Applications **1302** (which may alternatively be called software instances, virtual appliances, network functions, virtual nodes, virtual network functions, etc.) are run in the virtualization environment **1300** to implement some of the features, functions, and/or benefits of some of the embodiments disclosed herein.

[0125] Hardware **1304** includes processing circuitry, memory that stores software and/or instructions executable by hardware processing circuitry, and/or other hardware devices as described herein, such as a network interface, input/output interface, and so forth. Software may be executed by the processing circuitry to instantiate one or more virtualization layers **1306** (also referred to as hypervisors or virtual machine monitors (VMMs)), provide VMs **1308a** and **1308b** (one or more of which may be generally referred to as VMs **1308**), and/or perform any of the functions, features and/or benefits described in relation with some embodiments described herein. The virtualization layer **1306** may present a virtual operating platform that appears like networking hardware to the VMs **1308**.

[0126] The VMs **1308** comprise virtual processing, virtual memory, virtual networking or interface and virtual storage, and may be run by a corresponding virtualization layer **1306**. Different embodiments of the instance of a virtual appliance **1302** may be implemented on one or more of VMs **1308**, and the implementations may be made in different ways. Virtualization of the hardware is in some contexts referred to as network function virtualization (NFV). NFV may be used to consolidate many network equipment types onto industry standard high volume server hardware, physical switches, and physical storage, which can be located in data centers, and customer premise equipment.

[0127] In the context of NFV, a VM **1308** may be a software implementation of a physical machine that runs programs as if they were executing on a physical, non-virtualized machine. Each of the VMs **1308**, and that part of hardware **1304** that executes that VM, be it hardware dedicated to that VM and/or hardware shared by that VM with others of the VMs, forms separate virtual network elements. Still in the context of NFV, a virtual network function is responsible for handling specific network functions that run in one or more VMs **1308** on top of the hardware **1304** and corresponds to the application **1302**.

[0128] Hardware **1304** may be implemented in a standalone network node with generic or specific components. Hardware **1304** may implement some functions via virtualization. Alternatively, hardware **1304** may be part of a larger cluster of hardware (e.g., such as in a data center or CPE) where many hardware nodes work together and are managed via management and orchestration **1310**, which, among others, oversees lifecycle management of applications **1302**. In some embodiments, hardware **1304** is coupled to one or more radio units that each include one or more transmitters and one or more receivers that may be coupled to one or more antennas. Radio units may communicate directly with other hardware nodes via one or more appropriate network interfaces and may be used in combination with the virtual components to provide a virtual node

with radio capabilities, such as a radio access node or a base station. In some embodiments, some signaling can be provided with the use of a control system **1312** which may alternatively be used for communication between hardware nodes and radio units.

[0129] FIG. **14** shows a communication diagram of a host **1402** communicating via a network node **1404** with a UE **1406** over a partially wireless connection in accordance with some embodiments. Example implementations, in accordance with various embodiments, of the UE (such as a UE **912a** of FIG. **9** and/or UE **1000** of FIG. **10**), network node (such as network node **910a** of FIG. **9** and/or network node **1100** of FIG. **11**), and host (such as host **916** of FIG. **9** and/or host **1200** of FIG. **12**) discussed in the preceding paragraphs will now be described with reference to FIG. **14**.

[0130] Like host **1200**, embodiments of host **1402** include hardware, such as a communication interface, processing circuitry, and memory. The host **1402** also includes software, which is stored in or accessible by the host **1402** and executable by the processing circuitry. The software includes a host application that may be operable to provide a service to a remote user, such as the UE **1406** connecting via an over-the-top (OTT) connection **1450** extending between the UE **1406** and host **1402**. In providing the service to the remote user, a host application may provide user data which is transmitted using the OTT connection **1450**.

[0131] The network node **1404** includes hardware enabling it to communicate with the host **1402** and UE **1406**. The connection **1460** may be direct or pass through a core network (like core network **906** of FIG. **9**) and/or one or more other intermediate networks, such as one or more public, private, or hosted networks. For example, an intermediate network may be a backbone network or the Internet.

[0132] The UE **1406** includes hardware and software, which is stored in or accessible by UE **1406** and executable by the UE's processing circuitry. The software includes a client application, such as a web browser or operator-specific "app" that may be operable to provide a service to a human or non-human user via UE **1406** with the support of the host **1402**. In the host **1402**, an executing host application may communicate with the executing client application via the OTT connection **1450** terminating at the UE **1406** and host **1402**. In providing the service to the user, the UE's client application may receive request data from the host's host application and provide user data in response to the request data. The OTT connection **1450** may transfer both the request data and the user data. The UE's client application may interact with the user to generate the user data that it provides to the host application through the OTT connection **1450**.

[0133] The OTT connection **1450** may extend via a connection **1460** between the host **1402** and the network node **1404** and via a wireless connection **1470** between the network node **1404** and the UE **1406** to provide the connection between the host **1402** and the UE **1406**. The connection **1460** and wireless connection **1470**, over which the OTT connection **1450** may be provided, have been drawn abstractly to illustrate the communication between the host **1402** and the UE **1406** via the network node **1404**, without explicit reference to any intermediary devices and the precise routing of messages via these devices.

[0134] As an example of transmitting data via the OTT connection **1450**, in step **1408**, the host **1402** provides user data, which may be performed by executing a host application. In some embodiments, the user data is associated with a particular human user interacting with the UE **1406**. In other embodiments, the user data is associated with a UE **1406** that shares data with the host **1402** without explicit human interaction. In step **1410**, the host **1402** initiates a transmission carrying the user data towards the UE **1406**. The host **1402** may initiate the transmission responsive to a request transmitted by the UE **1406**. The request may be caused by human interaction with the UE **1406** or by operation of the client application executing on the UE **1406**. The transmission may pass via the network node **1404**, in accordance with the teachings of the embodiments described throughout this disclosure. Accordingly, in step **1412**, the network node **1404** transmits to the UE **1406** the user data that was carried in the transmission that the host **1402** initiated, in accordance with the teachings of the embodiments described throughout this disclosure. In step **1414**, the UE

1406 receives the user data carried in the transmission, which may be performed by a client application executed on the UE **1406** associated with the host application executed by the host **1402**.

[0135] In some examples, the UE **1406** executes a client application which provides user data to the host **1402**. The user data may be provided in reaction or response to the data received from the host **1402**. Accordingly, in step **1416**, the UE **1406** may provide user data, which may be performed by executing the client application. In providing the user data, the client application may further consider user input received from the user via an input/output interface of the UE **1406**. Regardless of the specific manner in which the user data was provided, the UE **1406** initiates, in step **1418**, transmission of the user data towards the host **1402** via the network node **1404**. In step **1420**, in accordance with the teachings of the embodiments described throughout this disclosure, the network node **1404** receives user data from the UE **1406** and initiates transmission of the received user data towards the host **1402**. In step **1422**, the host **1402** receives the user data carried in the transmission initiated by the UE **1406**.

[0136] One or more of the various embodiments improve the performance of OTT services provided to the UE **1406** using the OTT connection **1450**, in which the wireless connection **1470** forms the last segment. More precisely, the teachings of these embodiments may reduce the risk of a medical patient's vital medical data being lost by the wireless network, and thereby going unnoticed by medical staff. Reducing this risk can improve patient care.

[0137] In an example scenario, factory status information may be collected and analyzed by the host **1402**. As another example, the host **1402** may process audio and video data which may have been retrieved from a UE for use in creating maps. As another example, the host **1402** may collect and analyze real-time data to assist in controlling vehicle congestion (e.g., controlling traffic lights). As another example, the host **1402** may store surveillance video uploaded by a UE. As another example, the host **1402** may store or control access to media content such as video, audio, VR or AR which it can broadcast, multicast or unicast to UEs. As other examples, the host **1402** may be used for energy pricing, remote control of non-time critical electrical load to balance power generation needs, location services, presentation services (such as compiling diagrams etc. from data collected from remote devices), or any other function of collecting, retrieving, storing, analyzing and/or transmitting data.

[0138] In some examples, a measurement procedure may be provided for the purpose of monitoring data rate, latency and other factors on which the one or more embodiments improve. There may further be an optional network functionality for reconfiguring the OTT connection **1450** between the host **1402** and UE **1406**, in response to variations in the measurement results. The measurement procedure and/or the network functionality for reconfiguring the OTT connection may be implemented in software and hardware of the host **1402** and/or UE **1406**. In some embodiments, sensors (not shown) may be deployed in or in association with other devices through which the OTT connection **1450** passes; the sensors may participate in the measurement procedure by supplying values of the monitored quantities exemplified above, or supplying values of other physical quantities from which software may compute or estimate the monitored quantities. The reconfiguring of the OTT connection **1450** may include message format, retransmission settings, preferred routing etc.; the reconfiguring need not directly alter the operation of the network node **1404**. Such procedures and functionalities may be known and practiced in the art. In certain embodiments, measurements may involve proprietary UE signaling that facilitates measurements of throughput, propagation times, latency and the like, by the host **1402**. The measurements may be implemented in that software causes messages to be transmitted, in particular empty or 'dummy' messages, using the OTT connection **1450** while monitoring propagation times, errors, etc.

[0139] Although the computing devices described herein (e.g., UEs, network nodes, hosts) may include the illustrated combination of hardware components, other embodiments may comprise computing devices with different combinations of components. It is to be understood that these

computing devices may comprise any suitable combination of hardware and/or software needed to perform the tasks, features, functions and methods disclosed herein. Determining, calculating, obtaining or similar operations described herein may be performed by processing circuitry, which may process information by, for example, converting the obtained information into other information, comparing the obtained information or converted information to information stored in the network node, and/or performing one or more operations based on the obtained information or converted information, and as a result of said processing making a determination. Moreover, while components are depicted as single boxes located within a larger box, or nested within multiple boxes, in practice, computing devices may comprise multiple different physical components that make up a single illustrated component, and functionality may be partitioned between separate components. For example, a communication interface may be configured to include any of the components described herein, and/or the functionality of the components may be partitioned between the processing circuitry and the communication interface. In another example, non-computationally intensive functions of any of such components may be implemented in software or firmware and computationally intensive functions may be implemented in hardware.

[0140] In certain embodiments, some or all of the functionality described herein may be provided by processing circuitry executing instructions stored on in memory, which in certain embodiments may be a computer program product in the form of a non-transitory computer-readable storage medium. In alternative embodiments, some or all of the functionality may be provided by the processing circuitry without executing instructions stored on a separate or discrete device-readable storage medium, such as in a hard-wired manner. In any of those particular embodiments, whether executing instructions stored on a non-transitory computer-readable storage medium or not, the processing circuitry can be configured to perform the described functionality. The benefits provided by such functionality are not limited to the processing circuitry alone or to other components of the computing device, but are enjoyed by the computing device as a whole, and/or by end users and a wireless network generally.

Claims

1. A method of operating a communication device in a communications network that includes a network node, the method comprising: determining information associated with a medical state of a patient associated with the communication device; determining a priority of data associated with the patient based on the information; and communicating data associated with the patient using the priority.
2. The method of claim 1, wherein determining the priority of the data associated with the patient based on the information comprises: transmitting the information to the network node; and responsive to transmitting the information, receiving an indication of a priority of the data.
3. The method of claim 2, wherein receiving the indication of the priority of the data comprises receiving a Quality of Service Flow Identifier, QFI, associated with a Quality of Service, QoS, profile associated with the data, and wherein communicating the data associated with the patient comprises transmitting a message comprising the data and the QFI in the header.
4. The method of claim 3, wherein the QoS profile comprises an indication of at least one of: an allocation and retention priority, ARP; a guaranteed flow bit rate, GFBR; a maximum flow bit rate, MFBR; a maximum packet loss rate; a delay critical resource type; and a notification control.
5. The method of claim 1, wherein determining the priority of the data associated with the patient based on the information comprises determining at least one of: whether the medical state of the patient exceeds a first threshold level; whether the medical state of the patient is below a second threshold level; whether the medical state of the patient is outside of a threshold range; and whether a rate of change in the medical state of the patient exceeds a threshold rate.
6. The method of claim 1, wherein the patient comprises a first patient of a plurality of patients

associated with the communication device, wherein communicating the data associated with the patient comprises: prioritizing the data associated with the first patient over data associated with a second patient of the plurality of patients.

7. The method of claim 1, wherein the information comprises at least one of: a respiration rate of the patient; an oxygen saturation of the patient; a systolic blood pressure of the patient; a pulse rate of the patient; a level of consciousness of the patient; a temperature of the patient; an aggregate numerical value indicating a medical state of the patient; and a term indicating a severity of the medical state of the patient.

8. The method of claim 1, wherein the network node comprises at least one of: a radio access network, RAN, node; and a core network, CN, node, and wherein the communication device comprises at least one of: a healthcare user equipment, HUE; a wearable device; and a medical sensor.

9. A method of operating a network node in a communications network that includes a communication device, the method comprising: receiving information associated with a medical state of a patient, the patient being associated with the communication device; determining a priority of data associated with the patient based on the information; and prioritizing data associated with the patient based on the priority of the data.

10. The method of claim 9, wherein receiving the information comprises receiving a request from the communication device to determine the priority of the data associated with the patient.

11. The method of claim 9, wherein receiving the information comprises receiving at least one of: a respiration rate of the patient; an oxygen saturation of the patient; a systolic blood pressure of the patient; a pulse rate of the patient; a level of consciousness of the patient; a temperature of the patient; an aggregate numerical value indicating a medical state of the patient; and a term indicating a severity of the medical state of the patient.

12. The method of claim 9, wherein determining the priority of the data associated with the patient comprises determining the priority of the data based on the medical state exceeding a threshold.

13. The method of claim 9, wherein the information comprises current information indicating a current medical state of the patient, wherein determining the priority of the data associated with the patient comprises determining the priority of the data based on a change between the current information and previous information indicating a previous medical state of the patient.

14. The method of claim 13, wherein determining the priority of the data associated with patient further comprises determining the priority of the data associated with the patient based on an amount of time between receiving the current information and receiving the previous information.

15. The method of claim 9, wherein determining the priority of the data comprises creating or modifying a Quality of Service, QoS, profile associated with the data.

16. The method of claim 15, wherein the QoS profile comprises an indication of at least one of: an allocation and retention priority, ARP; a guaranteed flow bit rate, GFBR; a maximum flow bit rate, MFBR; a maximum packet loss rate; a delay critical resource type; and a notification control.

17. The method of claim 15, wherein prioritizing the data associated with the patient comprises: transmitting an indication of a QoS Flow Identifier, QFI, associated with the QoS profile to the communication device with instructions to include the QFI in a header of messages that include the data.

18. The method of claim 9, wherein prioritizing the data associated with the patient comprises: receiving a plurality of data including data associated with the patient; and communicating the data associated with the patient before other data of the plurality of data based on the priority of the data associated with the patient.

19. The method of claim 9, wherein prioritizing the data associated with the patient comprises: allocating more network resources for the data associated with the patient than for other data being communicated on the communications network.

20. (canceled)

21. A communication device, the communication device comprising: processing circuitry; and memory coupled to the processing circuitry and having instructions stored therein that are executable by the processing circuitry to cause the communication device to perform operations comprising: determining information associated with a medical state of a patient associated with the communication device; determining a priority of data associated with the patient based on the information; and communicating data associated with the patient using the priority.

22.-24. (canceled)
